

NEBB TAB Certification Exam Abbreviations

Nomenclatures: Air / Fan Equations	Nomenclatures: Hydraulics / Pump Equations
A = Area (sq. ft.)	Cv = Valve Constant
bhp = Brake Horsepower	D = Impeller Diameter
Btu = British Thermal Unit	EP = Pump Efficiency
Btu/h = Btuh = BTUH = BTU/Hour	f = Friction Factor
C = Friction Loss Coefficient (For Duct Fittings)	ft = Foot
C = Center Distance (used in belt formula)	g = Acceleration of Gravity (32 ft/sec ²)
CFM = Cubic Feet Per Minute	gal = Gallons
Cp = Specific Heat (BTU/lb°F)	GPM = Gallons Per Minute
d = Density (lb/ft ³)	H = Head (ft w.c.)
d = PD _{motor} = Motor Sheave Pitch Diameter	h = Head Loss (ft w.c.)
D = PD _{fan} = Fan Sheave Pitch Diameter	Hg. = Mercury
Δ = Difference or Change (Final - Initial)	L = Length
ft = Foot	lb = Pounds
FPM = Feet per Minute	NPSHA = Net Positive Suction Head Available (ft)
h = Enthalpy (BTU/lb dry air)	P = Pressure
Δh = Change in Enthalpy	ΔP = Pressure Difference
hr = Hour	Pa = Atmospheric Pressure (34 ft. w.g.)
H = Height	Ps = Pressure at Pump Centerline (ft. w.g.)
kW = Kilowatt	Pvp = Absolute Vapor Pressure (ft. w.g.)
L = Length	psi = Pounds Per Square Inch
M = Mass	RPM = Revolutions per minute
P = Pressure (Static or Total)	Sp. Gr. = Specific Gravity (for water use 1.00)
Pa = Absolute Pressure (Atmospheric Pressure + Gauge Pressure)	TS = Tip Speed
P _b = Absolute Static Pressure (inches Hg.)	V = Velocity (ft/min)
Pi = 3.14 or π	V ² ÷ 2g = Velocity head at point P _a (ft. w.g.)
psia = Pounds Per Square Inch Absolute	WHP = Water Horsepower
psig = Pounds Per Square Inch Gauge	w.g. = w.c. = water gauge = water column
Q = Heat Flow Rate (BTU/Hour)	
R = Sum of Thermal Resistances (ft ² · hr · °F/Btu)	Nomenclatures: Temperatures
R = Gas Constant (ft ² /R)	°C = Degrees Celsius
R = Radius	°F = Degrees Fahrenheit
RPM = Revolutions Per Minute	h _{mix} = Mixed Air Enthalpy
SP = Static Pressure	h _{r/a} = Return Air Enthalpy
t = Temperature (°F)	h _{o/a} = Outside Air Enthalpy
T = Absolute Temperature (460° + t) or °R	K = Kelvin
TP = Total Pressure (SP + VP)	% o/a = % of Outside Air
U = Heat Transfer Coefficient (BTU/ft ² ·hr·°F)	% r/a = % of Return Air
V = Velocity (ft/min)	°R = Degrees Rankine
V = Volume (ft ³)	T _{mix} = Mixed Air Temperature
VM = Measured Velocity (ft/min)	T _{r/a} = Return Air Temperature
VP = Vp = Velocity Pressure (in.w.g.)	T _{o/a} = Outside Air Temperature
W = Width	
ω = Humidity Ratio (lb or grains of water/lb of dry air)	
Δω = Change in Humidity Ratio	
Nomenclatures: Electrical	
A = Amps (I)	
BHP = Brake Horsepower	
E = Volts (V)	
FLA = Full Load Amps	
Hp = Horsepower	
I = Amps (A)	
kW = Kilowatt	
MO = Actual Motor Operating Amps	
NL = Motor No Load Amps	
Ω = Ohms (Resistance)	
P = Power (watts)	
P.F. = Power Factor (cos(°))	
R = Resistance (ohms, Ω)	
V = Volts (E)	
W = Watts or Power	
Eff. = Efficiency (of Motor or Pump)	



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NEBB Formulas and Definitions (Input English Units Only)

Temperature Formulas		Fan Laws		Hydronic Formulas	
$^{\circ}\text{F} = 1.8\text{C} + 32^{\circ}$	$^{\circ}\text{C} = (^{\circ}\text{F} - 32) / 1.8$	$\text{CFM}_2 \div \text{CFM}_1 = \text{RPM}_2 \div \text{RPM}_1$		$\text{Ft w.c.} = \text{PSI} \times 2.31$	$\text{PSI} = \text{Ft w.c.} \div 2.31$
$^{\circ}\text{R} = ^{\circ}\text{F} + 460$	$\text{K} = ^{\circ}\text{C} + 273$	$\text{RPM}_2 = (\text{CFM}_2 \div \text{CFM}_1) \times \text{RPM}_1$		$\text{Ft} = \text{Inches} \div 12$	$\text{Inches} = \text{Ft} \times 12$
Mixed Air Temperatures					
$T_{\text{mix}} = (\% \text{ o/a} \times T_{\text{o/a}}) + (\% \text{ r/a} \times T_{\text{r/a}})$		$\text{CFM}_2 = (\text{RPM}_2 \div \text{RPM}_1) \times \text{CFM}_1$		$1 \text{ ft w.c.} = 0.433 \text{ psi}$	$1 \text{ psi} = 2.31 \text{ ft. w.c.}$
$h_{\text{mix}} = (\% \text{ o/a} \times h_{\text{o/a}}) + (\% \text{ r/a} \times h_{\text{r/a}})$		$P_2 \div P_1 = (\text{CFM}_2 \div \text{CFM}_1)^2$		$1 \text{ ft Hg.} = 5.89 \text{ psi}$	$1 \text{ psi} = 2.04 \text{ in. Hg.}$
$\% \text{O/A} = \frac{(T_{\text{r/a}} - T_{\text{mix}})}{(T_{\text{r/a}} - T_{\text{o/a}})}$	$\% \text{O/A} = \frac{(h_{\text{r/a}} - h_{\text{mix}})}{(h_{\text{r/a}} - h_{\text{o/a}})}$	$P_2 = (\text{CFM}_2 \div \text{CFM}_1)^2 \times P_1$		$1 \text{ in. Hg.} = 13.6 \text{ in. w.g.}$	$1 \text{ in Hg.} = 1.13 \text{ ft.}$
Velocity (V) & CFM Formulas					
$Q = V \times A$	$\text{CFM} = V \times A$	$\text{CFM}_2 = \sqrt[3]{(P_2 \div P_1)} \times \text{CFM}_1$		$\text{Coil PD: } \text{GPM}_2 = \sqrt[3]{(P_2 \div P_1)} \times \text{GPM}_1$	
$\text{CFM} = \text{Velocity (FPM)} \times \text{Area (sq.ft)}$		$\text{BHP}_2 \div \text{BHP}_1 = (\text{CFM}_2 \div \text{CFM}_1)^3$		$\text{Cv} = \text{GPM} \div \sqrt[3]{\Delta P (\text{psi})}$	$\Delta P (\text{psi}) = (\text{GPM} \div \text{Cv})^2$
$\text{TP} = \text{VP} + \text{SP}$		$\text{BHP}_2 = (\text{CFM}_2 \div \text{CFM}_1)^3 \times \text{BHP}_1$		$\text{GPM} = \text{Cv} \times \sqrt[3]{\Delta P (\text{psi})}$	
$\text{VP} = \text{TP} - \text{SP}$	$\text{SP} = \text{TP} - \text{VP}$	$\text{CFM}_2 = \sqrt[3]{(\text{BHP}_2 \div \text{BHP}_1)} \times \text{CFM}_1$		$\text{WHP} = \text{GPM} \times \text{Head (ft.)} \times \text{Sp.Gr} / 3960$	
$\text{TP} = \text{C} \times \text{VP}$, where C=duct fitting loss coef.		Tip Speed = $(\text{Pi} \times \text{D} \times \text{RPM}) \div 12$ or		$\text{BHP} = \text{GPM} \times \text{Head (ft.)} \times \text{Sp.Gr.} / (3960 \times E_p)$	
$\text{VP} = \text{TP} - \text{SP}$		$\text{RPM} = (\text{Tip Speed} \times 12) \div (\text{Pi} \times \text{D})$		$\text{Sp.Gr.} = 1.0$, unless given	
		Electrical			
$\text{Vp} = (\text{V} \div 4005)^2$	$\text{V} = 4005 \times \sqrt[2]{\text{Vp}}$	$1 \text{ watt} = 3.413 \text{ Btu/h}$	$\text{Btu/h} = \text{Watt} \times 3.413$	$\text{Eff of a pump} = \text{whp} \times 100 \div \text{bhp}$	
$\text{Air Changes per hour} = \text{CFM} \times 60 \div (\text{L} \times \text{W} \times \text{H})$		$\text{kW} = \text{Btu/h} \div 3413$	$\text{Btu/h} = \text{kW} \times 3413$	$\text{NPSHA} = P_a \div P_s + (\text{V}^2 \div 2g) - P_{vp} - P_f$	
$\text{Area of Round Duct} = \pi r^2$		$\text{Power (P)} = \text{V} \times \text{A}$	$\text{P} = \text{E} \times \text{I}$	$\text{H} = f \times \text{L} \div \text{D} \times (\text{V}^2 \div 2g)$	
$\text{Duct Diameter} = \sqrt[2]{[(\text{Area} \times 144) \div \text{Pi}]} \times 2$		$\text{Watts} = \text{kW} \times 1000$	$\text{V} = \text{A} \times \Omega$ or $\text{E} = \text{I} \times \text{R}$	Pump Laws	
$\text{Area of Oval (ft}^2\text{)} = (\text{W}_{\text{rect}} (\text{in.}) \times \text{H} (\text{in.}) + 0.25 \times \text{Pi} \times \text{H}^2) \div 144$		$\text{kW} = \text{Watts} \div 1000$	$\text{A} = \text{V} \div \Omega$ or $\text{I} = \text{E} \div \text{R}$	$\text{GPM}_2 \div \text{GPM}_1 = \text{RPM}_2 \div \text{RPM}_1$	
$\text{psia} = \text{psig} + 14.7$	$\text{psig} = \text{psia} - 14.7$	$\text{Watts} = \text{V} \times \text{A}$	$\Omega = \text{V} \div \text{A}$ or $\text{R} = \text{E} \div \text{I}$	$\text{RPM}_2 = (\text{GPM}_2 \div \text{GPM}_1) \times \text{RPM}_1$	
$\text{V} = 1096 \times \sqrt[2]{(\text{Vp} \div d)}$; $d = 1.325 (\text{Pb/T})$ or Std air $d = 0.75$		$\text{kW} = \text{V} \times \text{A} \div 1000$ (1 ph)	$\text{kW} = (\text{V} \times \text{A} \times 1.73) \div 1000$ (3 ph)	$\text{GPM}_2 = (\text{RPM}_2 \div \text{RPM}_1) \times \text{GPM}_1$	
Btu Formula Q = Heat Transfer Rate = Heat Flow (Btu/h)		$\text{V} = (\text{kW} \times 1000) \div \text{A}$	$\text{A} = (\text{kW} \times 1000) \div \text{V}$	$\text{GPM}_2 \div \text{GPM}_1 = \text{D}_2 \div \text{D}_1$	
$\text{BTU (total)} = \text{BTU (latent)} + \text{BTU (sensible)}$		Series Circuit: total of resistance: $\Omega = \text{R}_{\text{total}}$		$\text{D}_2 = (\text{GPM}_2 \div \text{GPM}_1) \times \text{D}_1$	
$\text{Q (total)} = 500 \times \text{GPM} \times \Delta \text{T of water}$		$\text{R}_{\text{total}} = \text{R}_1 + \text{R}_2 + \text{R}_3 \dots$		$\text{GPM}_2 = (\text{D}_2 \div \text{D}_1) \times \text{GPM}_1$	
$\text{Q (total)} = 4.5 \times \text{CFM} \times \Delta \text{h}$		Parallel Circuit: total of resistance: $\Omega = \text{R}_{\text{total}}$		$\text{H}_2 \div \text{H}_1 = (\text{GPM}_2 \div \text{GPM}_1)^2$	
$\text{Q (sensible)} = 1.08 \times \text{CFM} \times \Delta \text{T}$		$1/\text{R}_{\text{total}} = 1/\text{R}_1 + 1/\text{R}_2 + 1/\text{R}_3 \dots$		$\text{H}_2 = (\text{GPM}_2 \div \text{GPM}_1)^2 \times \text{H}_1$	
$\text{Q (sensible)} = 60 \times \text{Cp} \times d \times \text{CFM} \times \Delta \text{T}$		Voltage Drop = $\text{A} \times \text{R}_{\text{total}}$		$\text{GPM}_2 = \sqrt[2]{(\text{H}_2 \div \text{H}_1)} \times \text{GPM}_1$	
for Standard air: $\text{Cp} = 0.24 \text{ Btu/lb} \times ^{\circ}\text{F}$		$\text{V}_{\text{primary}} \times \text{A}_{\text{primary}} = \text{V}_{\text{secondary}} \times \text{A}_{\text{secondary}}$		$\text{BHP}_2 \div \text{BHP}_1 = (\text{GPM}_2 \div \text{GPM}_1)^3$	
$\text{Q (latent)} = 0.69 \times \text{CFM} \times \Delta \text{W (gr.)}$		Brake Horse Power		$\text{BHP}_2 = (\text{GPM}_2 \div \text{GPM}_1)^3 \times \text{BHP}_1$	
$\text{Q (latent)} = 4840 \times \text{CFM} \times \Delta \text{W (lb.)}$		$1 \text{ HP} = 746 \text{ Watts}$	$1 \text{ HP} = 0.746 \text{ kW}$	$\text{GPM}_2 = \sqrt[3]{(\text{BHP}_2 \div \text{BHP}_1)} \times \text{GPM}_1$	
$\Delta \text{T} = \text{Sens BTUs} \div (1.08 \times \text{CFM})$		$\text{BHP}_{(1\text{ph})} = (\text{V} \times \text{A} \times \text{P.F.} \times \text{Eff}) \div 746$		Boiler Formulas	
$\text{Btu/h} = \text{kW} \times 3412$	$1 \text{ ton refriger.} = 12,000 \text{ Btu/h}$	$\text{BHP}_{(3\text{ph})} = (\text{V} \times \text{A} \times \text{P.F.} \times \text{Eff} \times 1.732) \div 746$		$\text{Output BTU} = \text{Input BTU} \times \% \text{ Eff}$	
$\text{Btu} = 1.06 \times \text{kJ}$		$\text{P.F.} = 0.8$ and $\text{Eff.} = 0.9$ if not given		$\text{Fire Rate} = \text{Input BTU} \div \text{Fuel BTU (gal / hr)}$	
$\Delta \text{T} = (\text{kW} \times 3413) \div (1.08 \times \text{CFM})$		$\text{BHP} = \text{HP} \times (\text{Amps}_{\text{actual}} - (\text{NL Amps} \times 0.5))$		$\text{Boiler Operating Cost} =$	
$\text{Latent Heat of Vaporization} = 970 \text{ Btu/lb}$		$(\text{Actual FLA}) - (\text{NL Amps} \times 0.5)$		$\text{Fire Rate(gph)} \times \text{Hrs Burned} \times \Cost/gal	
$\text{Latent Heat of Fusion} = 144 \text{ Btu/lb}$		$\text{Actual FLA} = (\text{FLA}_{\text{tag}} \times \text{Volts}_{\text{tag}}) \div \text{Actual Volts}$		$\text{BTU} = \text{lbs of water} \times 500 \times \Delta \text{T}_{\text{water}}$	
		$\text{Fan HP} = (\text{CFM} \times \text{TP} \times \text{SG}) \div (6356 \times \text{eff.})$			
Sheaves: PD= Pitch Diameter, C = Center Distance		Conversions / Definitions @ Standard Temperature and Pressure = 14.69 psia at 68 °F			
$\text{RPM}_{\text{fan}} \div \text{RPM}_{\text{motor}} = \text{PD}_{\text{motor}} \div \text{PD}_{\text{fan}}$		$\text{Specific Heat (Cp) water} = 1.00 \text{ Btu/lb} \times ^{\circ}\text{F} @ 68\text{F}$		$\text{Specific Volumn of Air (Std)} = 13.33 \text{ Ft}^3/\text{lb}$	
$\text{RPM}_{\text{fan}} = (\text{PD}_{\text{motor}} \div \text{PD}_{\text{fan}}) \times \text{RPM}_{\text{motor}}$		$\text{Specific Heat(Cp) vapor} = 0.45 \text{ Btu/lb} \times ^{\circ}\text{F} @ 68\text{F}$		$\text{Atmosph. Press} = 29.92 \text{ in.Hg or } 14.69 \text{ psi}$	
$\text{RPM}_{\text{motor}} = (\text{PD}_{\text{fan}} \div \text{PD}_{\text{motor}}) \times \text{RPM}_{\text{fan}}$		$\text{Specific Heat(Cp) of dry air} = 0.24 \text{ Btu/lb} \times ^{\circ}\text{F} @ 68\text{F}$		$\text{Density of Air (Std)} = 0.075 \text{ lb/Ft}^3$	
$\text{PD}_{\text{motor}} = (\text{RPM}_{\text{fan}} \div \text{RPM}_{\text{motor}}) \times \text{PD}_{\text{fan}}$		$\text{Specific Heat (Cp) ice} = 0.50 \text{ Btu/lb} \times ^{\circ}\text{F} @ 68\text{F}$		$7000 \text{ grains} = 1 \text{ lb}$	
$\text{PD}_{\text{fan}} = (\text{RPM}_{\text{motor}} \div \text{RPM}_{\text{fan}}) \times \text{PD}_{\text{motor}}$		$1 \text{ Btu} = \text{Energy required to heat } 1 \text{ lb of water from } 39^{\circ}\text{F to } 40^{\circ}\text{F}$			
$\text{Belt Length} = 2\text{C} + 1.57 \times (\text{D} + \text{d}) + ((\text{D}-\text{d})^2 \div 4\text{C})$		$\text{Density of Water (Std)} = 62.4 \text{ lb/ft}^3$ or 7.49 gal/ft^3 or 8.33 lb/gal			
		Thermal Resistance			
		$\text{Q} = \text{U} \times \text{A} \times \Delta \text{T}$	$\text{R} = 1/\text{U}$		

Air Density Correction Factors (U.S. Units) Standard Air Density (Sea Level and 70°F) = 0.075 lb/ft ³ at 29.92 in. Hg.												
Altitude (Ft.)	Sea Level	1000	2000	3000	4000	5000	6000	7000	8000	9000	10000	
Barometer (in. of Hg.)	29.92	28.86	27.82	26.82	25.84	24.9	23.98	23.09	22.22	21.39	20.58	
Barometer (in. of w.g.)	407.5	392.8	378.6	365.0	351.7	333.9	326.4	314.8	302.1	291.1	280.1	
Air Temp °F	-40°	1.26	1.22	1.17	1.13	1.09	1.05	1.01	0.97	0.93	0.90	0.87
	0°	1.15	1.11	1.07	1.03	0.99	0.95	0.91	0.89	0.85	0.82	0.79
	40°	1.06	1.02	0.99	0.95	0.92	0.88	0.85	0.82	0.79	0.76	0.73
	70°	1.00	0.96	0.93	0.89	0.86	0.83	0.80	0.77	0.74	0.71	0.69
	100°	0.95	0.92	0.88	0.85	0.81	0.78	0.75	0.73	0.70	0.68	0.65
	150°	0.87	0.84	0.81	0.78	0.75	0.72	0.69	0.67	0.65	0.62	0.60
	200°	0.80	0.77	0.74	0.71	0.69	0.66	0.64	0.62	0.60	0.57	0.55
	250°	0.75	0.72	0.70	0.67	0.64	0.62	0.60	0.58	0.56	0.54	0.51
	300°	0.70	0.67	0.65	0.62	0.60	0.58	0.56	0.54	0.52	0.50	0.48
	350°	0.65	0.62	0.60	0.58	0.56	0.54	0.52	0.51	0.49	0.47	0.45
	400°	0.62	0.60	0.57	0.55	0.53	0.51	0.49	0.48	0.46	0.44	0.42
	450°	0.55	0.56	0.54	0.52	0.50	0.48	0.46	0.45	0.43	0.42	0.40
$V_2 \div V_1 = P_2 \div P_1$ $V_2 = (V_1 \times P_1) \div P_2$ V = Volume of Air P = Absolute Pressure	500°	0.54	0.53	0.51	0.49	0.47	0.45	0.44	0.43	0.41	0.39	0.38
	550°	0.53	0.51	0.49	0.47	0.45	0.44	0.42	0.41	0.39	0.38	0.36
	600°	0.50	0.48	0.46	0.45	0.43	0.41	0.40	0.39	0.37	0.35	0.34
	700°	0.46	0.44	0.43	0.41	0.39	0.38	0.37	0.35	0.34	0.33	0.32
	800°	0.42	0.40	0.39	0.37	0.36	0.35	0.33	0.32	0.31	0.30	0.29

Water Temperature °F	60°	150°	200°	250°	300°	340°
Feet Head differential per inch Hg. Differential	1.046	1.07	1.09	1.11	1.15	1.165

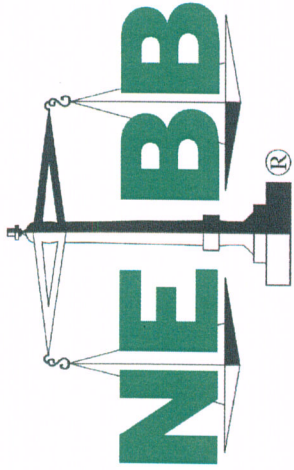
Duct Leakage Table	TABLE 3-1		
Duct SP in.w.g.	0.5, 1, 2"	3"	4,6,10"
Seal Class	C	B	A
	Duct Leak Class CFM/100 sq.ft.		
Rect Metal	24	12	6
Round Metal	12	6	3
Rect Fibr..	6	n/a	n/a
Round Fibr..	3	n/a	n/a

Allowable CFM Leakage = Sq.Ft. of Duct x Duct Leak Class (table 3-1) ÷ 100 sq.ft.
% Leakage = Allowable CFM Leakage ÷ Design CFM



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PSYCHROMETRIC CHART NORMAL TEMPERATURE

I-P Units

SEA LEVEL

BAROMETRIC PRESSURE: 29.921 IN. OF MERCURY

